CHAPTER 3 km reanalysis — known properties of the archive

Third of the three documents that ship with this archive. `CHAPTER_VARIABLES.md` says what is in the files; `MISSING_VARIABLES.md` says what is not and why. This one answers the question neither of those asks: **what is in the files that will surprise you.** Everything below is a property of the data as delivered. None of it is a defect awaiting repair: the defects the audit found were repaired before the archive was produced, and they are not listed here. What is listed is the archive behaving correctly in ways a user comparing it against ERA5, against observations, or against their own expectations will otherwise discover alone — and, having discovered one, will reasonably stop trusting the rest. Every number here was measured, not estimated. They come from a variable-by-variable audit run in September 2026 over a **34-timestep sample** spanning both sampled years and all four seasons (the sample is listed in `docs/audit/sample.md`); where a comparison against ERA5 is quoted, our 3 km field was upscaled to the ERA5 grid first. Each entry links to the issue that holds the measurement and the method.

How to read this document

Section 0 is the short list: the things that change what you do, in one line each. The rest states each property in full, with its number and what to do about it. Two kinds of entry are mixed together deliberately, because a user does not care which is which:

- **properties of the run** — the model was configured a certain way and the data reflect it. Nothing can be done after the fact; the run would have to be repeated.
- **properties of the conversion** — a deliberate convention, chosen and documented. These could be changed, and the entry says what was chosen instead and why.

Which archive this describes

The GRIB2 archive, `grib_v3`, **246 messages per hourly file**, as produced by the converter from 2026-09-29 onward. (The six repairs below landed on 2026-09-25; one further change on 2026-09-29 moved the snow-emissivity threshold, which shifts `skt` over snow-covered land by about 0.05 K RMSE and is already reflected in the figures in section 4.) Files written before that date carry six defects that were repaired on that date — relative humidity, specific humidity, the column integrals, snowfall, the water masks and convective inhibition. They are listed in the repository's audit record (`docs/audit/re-verification.md`) and are **not** properties of this archive. If you hold files from before the rebuild, replace them rather than reconciling them. **Every accumulated field covers the hour ending at its validity time**, 00 UTC included, as in ERA5 (#70). Files written before 2026-10-07 carry their accumulations summed from 00 UTC of the day instead, with nothing at 00 UTC: the hour 23→00 UTC was in none of them. They are rewritten in place; a file is current when its accumulations' reference time is the previous hour. **The whole 00 UTC timestep comes from the previous day's run**, instantaneous fields and pressure levels included (#72); see §8.5. 00 UTC files written before 15 October 2026 took their instantaneous fields from the next run, six hours into its spin-up. They are rewritten in place, and nothing inside a file tells the two apart: replace any 00 UTC file obtained before that date.

0. The short list

#	Property	The number
1.1	Surface shortwave is high under clear sky: the run carried **no aerosol**	**+8.5 %** clear-sky, +13.6 % all-sky vs ERA5

#	Property	The number
1.2	Total cloud cover is low, high cloud lowest — it is `CLDFRA`, not the overlap	**0.78** and **0.61** of ERA5
1.3	Land skin temperature is cold, and the archive cannot say why	**−1.64 K** mean, **−3.24 K** at midday
2.1	`fsr` is WRF's **local** roughness, ERA5's is **effective**	**0.32–0.48** of ERA5 over land
2.2	`zust` over land responds to resolved terrain	**1.6–2.1×** ERA5 over land, 1.1× over sea
3.1	**`r` and `2r` saturate differently, and each is right**	mixed phase vs liquid water
3.2	`r` above 100 % is normal and is not clipped	up to **139.7 %** at 300 hPa
3.3	Humidity rebuilt from `2t` and `2d` overshoots at saturation	peaks at **100.09 %**
4.1	Below ground the pressure levels are **clamped**, not extrapolated — ERA5 extrapolates	affects **47.6 %** of the domain at 1000 hPa
4.2	`z` is exactly `9.80665 × z_WRF`; WRF integrated with 9.81	**0.034 %** off hydrostatic balance
4.3	50 hPa is the model's highest level, **copied**, not interpolated	`z` low by **227 m** on average
5.1	`10fg` is the resolved 10 m wind, **not a gust parameterisation**	**0.625** of ERA5's gust
5.2	Its declared one-hour window is right **88 %** of the time	excess **0.499–0.784 m/s** on the bad hours
6.1	Three runoff variables carry one field's information	`ssro` non-zero at ≤ **19** points
6.2	`tcw` sums **six** species where ECMWF defines five	graupel residual to **26 kg/m²**
7.1	One masking rule: a point is missing where the ERA5 parameter is undefined	bitmaps, never zeros
7.2	`mucin` is **masked**, not zero-filled, where the scheme declined	missing on **71–96 %** of the domain
8.1	`generatingProcessIdentifier = 128` marks **every** message and discriminates nothing	246 of 246
8.2	`al` is a two-season table value by category, not a continuous field	8–11 distinct values domain-wide
8.3	`slor` is the slope of the **resolved** 3 km terrain, not of sub-grid orography	eccodes shows the ECMWF name

#	Property	The number
8.5	00 UTC is the **previous day's** model run, the whole timestep; the run switch falls between 00 and 01 UTC	`2t` jumps **1.8 K** RMS at the switch, 0.66 K in an ordinary hour

1. Radiation, cloud and the surface energy budget

These three belong together. Read any one of them alone and you will predict the other two wrongly.

1.1 Surface shortwave is 8.5 % high under clear sky, because the run carried no aerosol

RRTMG ran with a CAM ozone climatology and **no aerosol at all** (`aer_opt = 0`). ERA5 carries the Tegen aerosol climatology. Over this domain — Europe, the Mediterranean and the Saharan margin — an aerosol optical depth of 0.1 to 0.2 attenuates the direct beam by about the amount measured. Ratio of means against ERA5 over the sample:

field	ours / ERA5
`ssrdc` surface solar down, **clear sky**	**1.085**
`ssrd` surface solar down, all sky	**1.136**
`ssrc` surface net solar, clear sky	1.116
`ssr` surface net solar, all sky	1.178
`tsrc` top net solar, clear sky	1.031
`strd`, `strdc` downward thermal	0.989, 0.996
`ttr`, `ttrc` outgoing thermal	1.009, 0.990

The decomposition is the useful part: the clear-sky figure isolates the atmosphere with no cloud involved, and the thermal fields — untouched by aerosol — agree with ERA5 to within 1.1 %, which is the control that makes the attribution credible rather than a guess. **What to do.** If you use this archive for solar resource, for an energy application, or as training data against pyranometer observations, expect a systematic high bias in surface shortwave of about this size. It is a property of the run and cannot be removed after the fact. Measured in #45.

1.2 Total cloud cover runs 22 % below ERA5, and high cloud 39 % below

Sample means, upscaled to the ERA5 grid:

	ours	ERA5	ratio
`tcc`	0.381	0.487	**0.783**
`hcc`	0.169	0.276	**0.610**
`mcc`	0.161	0.205	0.786
`lcc`	0.231	0.268	0.862
`tciw`	0.0185	0.0166	**1.109**

The deficit holds at **every one of the 34 timesteps**, both years, every month. **It is not the overlap assumption, and that was checked first.** The converter uses maximum-random overlap where ECMWF uses exponential-random, which would yield more cover — the right direction. But random overlap is the *most* total cover any assumption can extract from a given profile, and computed point by point on our own `CLDFRA` the entire span of possible assumptions is only **2 to 4 % wide**, because `CLDFRA` is strongly bimodal. ERA5 is 28 % above us. The overlap

accounts for at most a seventh of the gap. The gap is in `CLDFRA` itself — WRF's Xu–Randall diagnostic under `ICLOUD = 1`. The sharpest statement of it: **we carry 11 % more cloud ice than ERA5 and diagnose 39 % less high cloud.** The condensate is there; the fraction the scheme assigns to it is not. **One caveat, stated rather than buried.** Part of the gap is resolution. Our `tcc` is overlapped at 3 km and then averaged to 31 km; ERA5's is overlapped on an already-smooth 31 km profile, which yields somewhat more cover. That effect is bounded by the same 2–4 % span, so it cannot account for 22 % — but it is real, and the figure should not be quoted as if it were all model error. Measured in #48.

1.3 Land skin temperature runs 1.6 K below ERA5, 3.2 K at midday — and this contradicts the two above

`skt` against ERA5, mean over the whole sample:

region	bias	correlation
sea	**+0.01 K**	0.95–0.98
land	**−1.64 K**, reaching **−3.24 K** at midday	0.95–0.98

At 00Z the land bias is −0.5 to −1.2 K; at 12Z it is −0.8 to −3.2 K. **The sea number is the control.** Over open water the model's skin temperature *is* the SST, and both reanalyses see nearly the same one. A bias of +0.01 K there says the derivation, the emissivity and the encoding are not the cause of anything (see 2.3): the land bias is the run. **And it pulls against §1.1 and §1.2.** More shortwave reaching the ground and less cloud overhead both put energy *into* the land surface, and the surface comes out colder — most so at the hour when the excess solar is largest. That is not a contradiction in the measurements; it is an unexplained partition, and reconciling it needs the surface energy budget. **`HFX`, `LH` and the ground heat flux are precisely the three fields this archive does not have** (see `MISSING_VARIABLES.md`), so it cannot be settled from the published data. Candidates that cannot be separated: the land-surface scheme partitioning more of the available energy into latent heat, a different soil-moisture climate, the resolved 3 km orography raising the mean elevation of an upscaled land cell, or a genuine cold bias. **What to do.** If you use `skt` over land for evaluation, bias correction, or ML training against satellite land surface temperature, apply the number. Do not predict a warm land surface from the radiation findings — the archive does the opposite. Measured in #57.

2. Surface exchange

2.1 `fsr` is a local roughness where ERA5's is effective

	ours	ERA5	ratio of medians
`fsr`, **land**	0.188–0.217 m	0.375–0.394 m	**0.32–0.48**
`fsr`, sea	0.0089–0.0091 m	0.0103–0.0105 m	1.5–1.9

ERA5's `fsr` is an **effective** roughness that carries the orographic drag the IFS parameterises at 31 km. WRF's `ZNT`, which is what this field is, is the **local** surface roughness of the land-use class — the orography is resolved at 3 km instead of parameterised. Over sea the two agree, because both are Charnock. `ZNT` is a land-use lookup with a snow override: one value per category — 0.70 m for the forest classes, 0.15 for savannas and grassland, 0.065 for barren, exactly 1.00 m on the 19 867 urban points — with a floor of 0.011 m where snow covers the surface. Nothing is capped or mis-scaled. **What to do.** Do not take `fsr` from this archive and `fsr` from ERA5 as the same quantity: over land they differ by a factor of two to three. A drag coefficient computed from ours and compared against one computed from ERA5's will show a discrepancy that is not physical. Measured in #56.

2.2 `zust` runs 1.6 to 2.1 times ERA5's over land, and agrees over sea

	ours	ERA5	ratio
`zust`, **land**	0.44–0.52 m/s	0.23–0.28 m/s	**1.62–2.06**
`zust`, sea	0.21–0.34 m/s	0.19–0.31 m/s	**1.08–1.12**, corr 0.85–0.90

This is the other half of §2.1's story, and the split is what identifies it: our roughness is a third of ERA5's while our friction velocity is twice it, and those cannot both be a scaling error on the same `z0`. At 3 km the terrain-driven wind is resolved rather than smoothed, and the friction velocity responds to it. The excess is largest at midday — 1.06–1.13 at 00Z against 1.40–1.75 at 12Z, in every region. The field is internally consistent with the published stresses: $|\tau| = \rho u^{*2}$ holds to 1.8e-6 N/m² and $zust = \sqrt{|\tau|/\rho}$ to 2.7e-4 m/s, at every point of every timestep checked. Measured in #56.

2.3 `skt` is derived, its accuracy is known, and it is seasonal

`TSK` was not written out by the run, so `skt` is inverted from the upward longwave flux: $LWUPB = \varepsilon \sigma T^4 + (1-\varepsilon)\, GLW$, with the per-category emissivity **measured from the run itself** rather than looked up in a table. The derivation is verified; what is seasonal is how well it reproduces the model's own state:

	figure
water gate — `skt` against the model's own SST	**0.0005 K RMSE**, absmax 0.002 K, ~910 000 points, every timestep
land points reproduced to 1e-4	**99.98 % in July … 93.05 % in January**
land `skt` RMSE from the emissivity table	**0.0012 K in July … 0.084–0.119 K in January and February**, p99 0.81 K, absmax 2.79 K

The whole of the seasonal spread is the snow rule: emissivity switches to a flat 0.98 under snow, and the switch is applied slightly earlier than the model's own transition completes. Over water, where the answer is known independently, the derivation is exact. **What to do.** Treat `skt` over water as the model's SST to a thousandth of a kelvin. Over land in winter, allow a tenth of a kelvin of derivation error on top of the physical bias of §1.3 — the two are unrelated and the derivation error (at most 0.119 K RMSE, in February) is more than ten times smaller. Measured in #55.

3. Humidity — read this before using `r` or `2r`

3.1 `r` and `2r` saturate against different phases, and each is right for what it is

This is the single most likely misreading of the archive, so it gets the plainest sentence available:

> **`r` (paramId 157, pressure levels) is relative humidity over the mixed phase. `2r` (paramId 260242, 2 m) is relative humidity over liquid water at every temperature. They are not the same quantity and will not agree at cold points.**

The reason is that each follows its own definition rather than a single house convention:

- **`r`** is ECMWF's paramId 157, which saturates over ice below 250.16 K, over water above 273.16 K, and quadratically between. The archive computes it from IFS Cy41r2's own equations (7.5, 7.6, 7.89, 7.90) with every constant taken from that cycle — the one ERA5 was produced with — so it is directly comparable with ERA5's `r`. Over ice the mixed-phase and over-water definitions differ by up to a factor **1.8**.

- `2r` follows the **WMO screen convention**, which saturates over liquid water at all temperatures. ERA5 publishes no 2 m relative humidity, so there is no ERA5 field to follow; the choice is the meteorological standard for a screen-level observation, which is what a 2 m humidity is compared against.

What to do. Do not build a profile that puts `2r` at the bottom of `r`. Do not compute a vertical humidity gradient across the two. If you need them on one convention, recompute the one you need from `q`, `t` and `sp` — all three are published. Decided and measured in #40.

3.2 `r` above 100 % is normal, not a defect

There is no clip. Supersaturation is physical, the model produces it, and ERA5 publishes it too — so the archive publishes it. Measured on a single file: **139.7 % at 300 hPa** in the ice regime, and **34 785 points above 100 %** at 400 hPa. Below ground, values reach further still — up to about 163 % — and that is the clamping of §4.1 rather than the humidity: the lowest model level's moisture, from 24 to 27 m above the terrain (§4.1), read at a nominal pressure of 1000 hPa. See §4.1 before treating it as a moisture error. If your ingest pipeline rejects relative humidity above 100, it will reject valid data from this archive. Measured in #40 and #44.

3.3 Humidity reconstructed from `2t` and `2d` peaks at 100.09 %

At saturated points `2d` exceeds `2t` by a fixed **0.0133 K**, on 6 000 to 64 000 points per timestep. Every violating point is at relative humidity 100 — it is the dewpoint inversion overshooting by a constant at exactly saturation, and the constant is the same number at every timestep in every season, which is what identifies it as arithmetic rather than noise. It was left in place deliberately. Clipping `2d` to `2t` would remove the visible crossings while leaving the same bias at every unsaturated point, put a kink in the field at RH 100, and break the exact consistency between `2d` and `Q2` that the audit verified at 3.6e-4 to 1.1e-2 in vapour pressure. That trades a constraint for the appearance of one. **What to do.** Expect a reconstruction that peaks a tenth of a per cent above saturation. If you need a hard bound, clip your own reconstruction rather than reading it as an inconsistency in the archive. Measured in #42.

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4. Vertical structure

4.1 Below ground the pressure levels are clamped to the lowest model level — ERA5 extrapolates

Where a pressure level lies below the terrain, this archive publishes **the lowest model level's value, verbatim**. It is not an extrapolation and not a fill: `t` matches the lowest model level to **exactly 0.0** at 100 % of below-ground points, `q` to 1e-9, `z` to 2.5e-4 m. There is a good half and a bad half, and both matter. **The good half.** The number is a real model state from about 24 to 27 m above the ground, not an invention. A user reading 1000 hPa over the Alps is reading near-surface air, which is a more defensible thing to publish than a synthetic lapse-rate extrapolation. It is also why the archive is coherent across the surface join: `2t − t(1000 hPa)` below ground is −0.9 K at midnight and +1.4 K at noon in July, which is a nocturnal inversion and a superadiabatic daytime surface layer over the ~22 m between the 2 m level and the lowest model level. **The bad half.** `z` then contradicts `sp`. Where the surface pressure says the 1000 hPa surface is underground, the published `z` at 1000 hPa puts it a median of **24 to 27 m above** the terrain. Both messages are in the same file. Anyone deriving a height from `z`, or a thickness between levels, meets that self-contradiction on **47.6 % of the domain at 1000 hPa** and 9 % at 925 hPa. **And ERA5 does not do this.** ERA5 extrapolates below ground with a lapse rate, so the two archives differ in *convention* exactly where 1000 hPa is underground — which is most of the land. This difference, not any defect, is the whole of the 1000 hPa discrepancy against ERA5: a `t` bias of −1.71 K at 1000 hPa against −0.05 K at 850 hPa, and a `z` correlation of 0.108. **What to do.** Mask the pressure levels against `sp` yourself if you need

a clean comparison against ERA5 or a thickness computation. Every field needed to do it is in the same file. Measured in #44.

4.2 `z` is exactly 9.80665 × WRF's height, and the run integrated with 9.81

Both halves of this are true and a user will find one of them alone:

- **By construction**, the published geopotential is standard gravity times WRF's own geopotential height. Divide `z` by 9.80665 and you recover the model's height in metres, exactly — at the surface and on every pressure level, with no residual.
- **WRF integrated the hydrostatic equation with 9.81.** So the same archive sits **0.034 %** off hydrostatic balance against its own `t` and `p`. That is 1 part in 2 900: about 2 m at 500 hPa, 7 m at 50 hPa.

Neither number can be removed without breaking the other. The convention chosen is the one that makes `z` a correct ECMWF geopotential and makes the surface and pressure-level geopotentials agree with each other exactly — which they do, verified point by point over 1 349 910 land points. Measured in #39.

4.3 50 hPa is the model's highest level, copied, not interpolated

The model top is at 50 hPa (`p_top_requested = 5000` Pa) and the highest model level, where temperature and geopotential live, sits at **51.9 hPa** on average. 50 hPa is therefore above the last level, and the interpolation does what it does above the model: it **copies** the highest level. Measured on 2024-01-19 00 UTC, `t` at 50 hPa equals the top level's temperature to 0.009 K and `z` at 50 hPa equals its geopotential to the metre. For `t` the copy costs little: between 51.9 and 50 hPa the stratosphere is close to isothermal. For `z` it does not: the hydrostatic thickness of that layer is **227 m** on average (214 to 239 m over the domain), so the published `z` at 50 hPa is **low by that amount**, about 5 % of the 100 → 50 hPa thickness. `u`, `v`, `q` and the other fields are likewise the 51.9 hPa values. **What to do.** Treat 50 hPa as the top of the model, not as an isobaric surface: do not use its `z` for a thickness or a geostrophic wind against 100 hPa without adding the missing layer. Nothing above 50 hPa exists in this run, at any level.

5. Wind

5.1 `10fg` is the strongest wind the model actually simulated, not an estimate of the gusts above it

What ERA5 does. A 31 km model cannot see a gust: the peak happens on scales of metres and seconds it never represents. ERA5 therefore *estimates* it — the mean 10 m wind, plus a term standing in for the turbulence the model has smoothed away, sized from the surface friction, plus, in convective conditions, a term proportional to the wind shear between 850 and 950 hPa (IFS Cy41r2 Part IV, §3.10.4, eq. 3.99). It is a calculated estimate of something the model does not resolve. **What CHAPTER does.** Nothing of the kind. This field is simply the highest 10 m wind speed the simulation itself produced during the hour. No turbulence term, no estimate — the model's own wind, at its own resolution, at its strongest moment of the hour.

comparison	**ratio**	**corr**
ours vs ERA5 `10fg` (gust scheme, hourly max)	**0.625**	0.813
ours vs ERA5 `i10fg` (gust scheme, instantaneous)	**0.644**	0.820
ours vs **our own instantaneous 10 m wind, same file**	**1.004–1.032**	—

The last row explains the first two. Taking the maximum over an hour raises the value by **half a per cent to three per cent** and no more — and it is below the instantaneous wind at zero points out of 2 220 273, which is the consistency check an hourly maximum must pass. So what is published under paramId 49 is, in practice, the 10 m wind speed. ERA5's gust runs at roughly 1.6 times its own mean wind. The offset is a definition, not weather: 0.593 to 0.673 across all 34 sample timesteps, both years, every month, every hour. **"But CHAPTER is a downscaling of ERA5 — shouldn't the two agree?"** It is a reasonable expectation, and it does not hold, for a specific reason: **a downscaling inherits the *flow*, not the *diagnostics*.** ERA5 supplies the boundary conditions — wind, temperature, humidity, pressure — and those are carried into the simulation. A gust is not among them. It is an output of the IFS's own boundary-layer post-processing, recomputed and discarded at every step, and it never enters the nested model. WRF then rebuilds everything from its own physics, and this run did not write out a gust diagnostic of its own. What is published here is the one gust-like field the run produced. There is a second half to this, and it pulls the other way. At 3 km, part of what ERA5 must parameterise is **resolved**: our friction velocity over land runs 1.6 to 2.1 times ERA5's (§2.2) precisely because the terrain-driven wind is computed rather than smoothed. So a 3 km model genuinely needs a *smaller* gust increment than a 31 km one — smaller, but not zero, and this archive applies zero. **What to do.** Reading `10fg` as a wind gust **under-predicts by about 38 %**, and no bias correction repairs that, because it is a different quantity rather than a mis-scaled one. The correlation of 0.81 means it ranks windy places and windy hours correctly, so it is usable for ranking, for patterns, and as a lower bound. It will not give you a gust threshold. Measured in #50.

5.2 And its declared one-hour window is right only 88 % of the time

The length of the window is not the difference. ERA5's hourly gust is also a maximum over the preceding hour, and our messages declare the same one: `stepType = max`, `stepRange = 0-1`, `typeOfStatisticalProcessing = 2`, reference time H−1. That encoding was verified correct. What the field does not always respect is its own label. **It is not always a one-hour maximum.** WRF resets `WSPD10MAX` at the history write, and measured over six whole days, 138 hourly transitions, **the reset was missed at 17 of them (12.3 %)**. Those hours carry a maximum over two hours or more under a message that declares one. How much it costs, as mean excess over the instantaneous 10 m wind of the same file:

	mean excess
hours that reset	**0.068–0.311 m/s**
hours that did not	**0.499–0.784 m/s**

Three to eight times larger — a 10 to 17 % overstatement on a domain mean of about 4.5 m/s, at hours whose message says one hour. **Worse, it moves those hours *closer* to ERA5's gust for the wrong reason**, so the error is not detectable from the value. It was left as `0-1` and stated here rather than repaired: detecting the carry-over requires the previous hour's wrfout, which is not on disk when a file is converted, and no intra-file detector exists — all seven of WRF's diagnostic maxima reset on the same alarm, so when it is missed they carry over together and none can witness for another. The hourly misses correlate with adaptive-timestep variability, but the mechanism was not established. **What to do.** Do not treat the `10fg` time window as exact when a single hour matters. Over a season or a distribution the effect is the 12 % contamination above, which is quantified; for a specific hour there is no flag in the file. Measured in #49.

6. Water cycle

6.1 Three runoff variables carry one field's information

timestep	sro mean (m)	ssro mean (m)	ssro non-zero points	ro mean (m)
2024-02-15T12	3.093e-5	**0**	**0 of 2 220 273**	3.093e-5
2024-07-15T12	2.347e-5	2.7e-8	**19**	2.350e-5
2024-07-15T18	1.064e-4	3.9e-8	**19**	1.065e-4
2024-08-15T12	1.453e-5	**0**	**0**	1.453e-5

The table gives each timestep's accumulation since 00 UTC, i.e. the sum of the published hourly values from 01 UTC to that hour: summing does not rescue ssro. It is identically zero on five of the eighteen non-00Z sample timesteps, and where it is not, its domain mean is four to eight orders of magnitude below sro. **ro equals sro to four significant figures at every timestep.** The identity ro = sro + ssro holds to 3.7e-9 m; there is simply nothing in the second term. The cause is the time reference, not the conversion. The model's sub-surface drainage counter is alive — non-zero at 462 to 8 363 points with maxima of 11 to 106 mm — but it barely moves *within a day*: on 2024-07-15 the same 1 647 points carry drainage at 00, 06, 12 and 18Z and only 19 of them grow. Each published ssro is the drainage of one hour, so it is almost always empty, and even its sum over a day is. Against ERA5 the partition is the opposite way round: sro at 1.0–4.6 of ERA5's, ssro at 0.000–0.004, ro at 0.19–0.85. **What to do.** Do not close a water budget on the assumption that this archive has a drainage term — it does not, at daily scale. ro and sro are, knowingly, duplicate messages. We are not short of surface runoff; we are missing the other half of the partition, and ERA5's dominant term is the one we do not carry. Measured in #51.

6.2 tcw sums six species where ECMWF defines five

ECMWF defines paramId 136 tcw as the sum of water vapour, liquid water, cloud ice, rain and snow — **five species**. This archive builds it from **six**, adding graupel, and there is no graupel column published separately because ERA5 has no parameter for one. The residual, tcw minus the five published components, against the independently computed graupel column:

timestep	residual (mean)	graupel column (mean)	residual max
2024-01-15T12	0.003525	0.003396	**5.69 kg/m²**
2024-04-15T12	0.010819	0.010863	**13.89 kg/m²**
2024-07-15T12	0.009762	0.010219	**26.46 kg/m²**
2024-10-15T12	0.007046	0.007267	**16.73 kg/m²**

In the domain mean it is 0.4 % of the condensate and invisible. **Inside a July convective cell it reaches 26 kg/m² — larger than the whole of the rest of the column's condensate.** Keeping graupel inside tcw was the deliberate choice: dropping it would make tcw match ECMWF's definition exactly and the closure identity hold, at the cost of the graupel mass leaving the archive altogether. **What to do.** If you reconstruct tcw from tcwv + tclw + tciw + tcrw + tcsw, you will not get tcw, and the difference is graupel. In convective conditions it is not small. Measured in #47.

7. Masks and missing values

7.1 One masking rule for the whole archive

A point is masked where the ERA5 parameter is not defined there — not where the model has nothing to say, and not where the number looks wrong. Masked points are a GRIB **bitmap**, never zeros, so a consumer can always tell "not defined here" from "zero here". Applied:

field	masked where
`sst`, `ci`	not sea
`cl` (zero) and `dl` (missing)	sea
`swvl1-4`, `stl1-4`	not **permanent** land
`tvl`	shrubland dominates — MODIS carries no shrub phenology and guessing was refused
`slt`	not permanent land
`mucin`	see §7.2

Two consequences are worth stating on their own. **The sea/inland-water distinction is not the land-use lake class.** WPS's MODIS class 21 calls the Black Sea, the Sea of Azov and part of the Baltic a lake — together 89 % of the cells this domain marks fully lake. Masking on it would have deleted `sst` from the Black Sea. The discriminator is connected-component **size**: the water components here are 844 431 cells (Atlantic + Mediterranean + Baltic + North Sea), 49 841 (Black Sea + Azov), 8 583 (Red Sea), then 1 225 (Vänern, the largest genuine inland lake) and smaller. The threshold sits in that gap with a factor of seven of slack. **The soil columns are masked off sea ice, and that discards something.** WRF reclassifies a water cell carrying sea ice as land, gives it soil type 16 and fills the column with water — `swvl` reaches exactly 1.000 against that soil type's porosity of 0.435. Those columns are the model's own state, but they are not soil, and ERA5 has no soil there because its land-sea mask is permanent. They are masked. Nothing is lost that cannot be recovered: `ci` is published, so the ice itself is in the file. Note the related fact that **`lsm` and `al` are *not* constant in time** in this archive — they follow the model hour by hour, because that is what it integrated, and WRF moves the land mask where sea ice appears. They are written in every message and must not be treated as one-per-archive statics. Decided and measured in #34, #36 and #38.

7.2 `mucin` is masked where the scheme declined, on most of the domain

The CAPE routine flags convective inhibition as missing wherever CAPE is below 100 J/kg. The archive writes a bitmap there rather than a zero, because zero would read as "nothing is stopping convection" at precisely the columns where the most is — a column with no CAPE is usually a column with a great deal of inhibition, and that is often *why* there is no CAPE. How much of the field is absent:

timestep	`mucin` present on	masked
2024-01-15T12	3.8 %	**96.2 %**
2024-02-15T12	5.7 %	94.3 %
2024-03-15T12	4.8 %	95.2 %
2024-07-15T12	**28.8 %**	71.2 %
2024-10-15T12	15.2 %	84.8 %
2025-02-15T12	5.5 %	94.5 %

On a winter day, 96 % of the `mucin` field is missing. Where we are silent, ERA5 carries an inhibition averaging 25 to 265 J/kg. And even where ours is present, it is 0.03 to 0.24 of ERA5's — because ours exists only in columns with CAPE ≥ 100, which are the least inhibited ones, while ERA5 reports the field more widely. **What to do.** Expect a sparse field, most sparse in winter. Do not fill the gaps with zero: that is the state the archive deliberately does not publish, and a model trained on it learns that low-CAPE situations have no convective inhibition, which is the inverse of the physics. `mucape` is dense and its zeros are real — no CAPE genuinely is no available energy. Decided and measured in #54.

8. Conventions that do not mean what they look like

8.1 `generatingProcessIdentifier = 128` marks every message in the file

Within `grib_v3` this key **discriminates nothing**: all 246 messages carry 128, because that is eccodes' own GRIB2 sample default. A consumer filtering on `generatingProcessIdentifier == 128` to find the accumulations selects the entire file, including every instantaneous field. The value's only real meaning is against the older GRIB1 archive, where 127 marks the run-init-referred messages. **What to do.** Test `stepType` / `typeOfStatisticalProcessing` and `stepRange`, which state it unambiguously: an accumulation has a one-hour step and the previous hour as reference time. See §3 of `CHAPTER_VARIABLES.md` for the time convention itself. Measured in #53.

8.2 `al` is a two-season table value by land-use category, not a continuous field

The run used no monthly satellite albedo (`usemonalb = .false.`), so the background albedo comes from the land-use table's WINTER and SUMMER columns by dominant category. Consequences:

- the field takes only **8 to 11 distinct values** over the whole domain;
- it changes at **1 171 937 land points — 89.9 % of the land mask — at the winter/summer boundary**, and every transition is a table row with no residue (0.20→0.17 croplands on 497 649 points, 0.23→0.25 barren on 369 219, and so on);
- SUMMER applies April to October and WINTER November to March, so on land the table part does not move within a day; only the sea-ice points, which follow the hourly land mask, can;
- off land it is 0.08 on water and **0.65 on sea ice**, the model's default ice albedo.

What to do. Do not read `al` as an observed or evolving albedo, and do not difference it across the seasonal boundary expecting a physical signal. Note also that it moves *twice* for another reason: at sea-ice points, following the land mask of §7.1. Measured in #37.

8.3 `slor` is the slope of the resolved terrain, and `sdor` is ERA5's quantity at 3 km

Two orography parameters, differing from ERA5 in two different ways:

- **`slor`** is published as the gradient magnitude of the **resolved** 3 km terrain, while paramId 163 is defined as the slope of the **sub-grid** orography. eccodes will show you the parameter's own name, *Slope of sub-gridscale orography*, and nothing in the message says otherwise — which is why it is stated here.
- **`sdor`** *is* the ERA5 quantity, evaluated on a 3 km cell instead of a 31 km one. The measured ratio of medians against ERA5 is 0.244, which is (3/31)^0.605 — a resolution rescaling, not a different definition.

Measured in #37.

8.4 "No ERA5 parameter fits" usually means "ERA5 does not publish it"

Several model fields are absent from this archive under a reason recorded as *no ERA5 parameter fits*. For most of them the accurate statement is different, and the distinction matters to anyone deciding whether to ask for the field:

field	the accurate reason
SNOALB	ECMWF defines **260161** **`mxsalb`**, maximum snow albedo — exactly this quantity. ERA5 does not publish it

field	the accurate reason
QGRAUP	ECMWF defines **260028 `grle`** and **260001 `tcolg`**. ERA5 does not publish them
REFL_10CM	ECMWF defines **260134 `bref`**. ERA5 does not publish it
LPI	ECMWF defines the **228050 `litoti`** family, unpublished by ERA5 — and LPI is a potential index, not a flash density, so it would not be `litoti` in any case
SNOWFALLAC	it is a geometric snow depth, not a water equivalent; ECMWF's **3066 `sde`** is the matching parameter and ERA5 does not publish it

The exclusions stand either way: this archive's publication rule is that a field ships only when the WRF quantity **is** the ERA5 one. But "no parameter exists" and "ERA5 does not publish it" are different claims, and only the second is true here. One field is worth naming separately. **`LAI` is in the model output and is not published.** CHAPTER_VARIABLES.md §6 explains at length why lai_lv/lai_hv are refused — the model carries one leaf area index per cell and ERA5 wants one per vegetation class, so any split would be our invention — without mentioning that the underlying field exists. It does. Note that `rdlai2d = .false.`, so it is a table lookup on the dominant category rather than an observed LAI, which is a second reason not to split it. Measured in #35.

8.5 00 UTC is the previous day's model run, and the run switch falls at 01 UTC

Each model run covers one day: it starts at 18 UTC of the day before and ends at 00 UTC of the day after, 30 hours. Its first six hours are spin-up and are never published. So the archive's day D is 01 to 23 UTC of the run for D, and its 00 UTC is the **last** hour of the run for D−1: every field of the file, instantaneous, on pressure levels and accumulated, comes from that one state. ERA5 takes its 00 UTC accumulations from the forecast started at 18 UTC for the same reason. The accumulations at 00 UTC are the hour 23→00 UTC of the run for D−1, its 24 h state minus its 23 UTC state. The hour is not small: on 2019-02-10 it held **3.1 %** of the day's precipitation, **7.6 %** of its snowfall and **4.2 %** of its downward longwave. The 24 hours from 01 UTC of D to 00 UTC of D+1 all come from the same run, so their sum is that run's own daily total. The run switch is still there, once a day, between 00 and 01 UTC. Its size is the difference between two runs at the same instant, not an hour of weather. Measured on 2024-01-18/19 over the full domain, root-mean-square:

	`2t` (K)	**`sp` (hPa)**	**10 m wind (m/s)**	**`t` 850 hPa (K)**	**`tcwv` (kg/m²)**
an hour within one run, 23→00 UTC	0.66	0.50	1.13	0.48	0.89
the switch, 00→01 UTC	**1.79**	**1.62**	**1.62**	**0.94**	**1.54**

The switch reaches 11.0 K in `2t` at single points. The hour 00→01 UTC is the run for D at lead 6→7, the first hour after its spin-up. **What to do.** Index on validity time. A tendency or a time derivative computed across 00→01 UTC sees the switch; on this day it is 2.7 times the hourly change in `2t`. Where the previous day's 24 h state is lost at the source, the **whole** 00 UTC timestep is missing (every message a bitmap), never filled from the next run. Measured in #70 and #72.

Where the measurements live

Every number in this document was produced by a script in `tools/audit/` of the CHAPTER repository and is written up, with its method, in `docs/audit/`:

document	covers
`docs/audit/sample.md`	the 34 timesteps and why they were chosen
`docs/audit/harness.md`	how a field is compared against ERA5
`docs/audit/f1-pressure-levels.md`	§3, §4
`docs/audit/f2-near-surface.md`	§3.3
`docs/audit/f3-column-integrals.md`	§1.2, §6.2
`docs/audit/f4-radiation.md`	§1.1
`docs/audit/f5-water-cycle.md`	§5, §6.1, §8.1, §8.5
`docs/audit/f6-soil-and-snow.md`	§7.1
`docs/audit/f7-static-and-surface.md`	§7.1, §8.2, §8.3
`docs/audit/f8-surface-exchange.md`	§1.3, §2, §7.2
`docs/audit/re-verification.md`	the gate the repaired converter passed before this archive was produced

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